

EDUCATION

Sep. 2021 – Present	Doctor of Philosophy in Computer Science, Hong Kong Polytechnic University Supervisor: Dr. Jieming Shi
Sep. 2017 – Jul. 2020	Master of Engineering in Computer Technology, Sichuan University Supervisor: Prof. Min Zhu Avg Score: 84.00 / 100 GPA: 3.45 / 4.0 Rank: Top-20
Sep. 2013 – Jul. 2017	Bachelor of Engineering in Computer Science and Technology, Sichuan University Avg Score: 87.52 / 100 GPA: 3.52 / 4.0 Rank: 24th out of 361 First Class Honors

RESEARCH INTERESTS

Spatial and Temporal Data Management, Urban Computing, Geo-spatial Data Analysis and Visualization

PUBLICATIONS

- [1] **Wei Tian**, Jieming Shi, Man Lung Yiu. [Efficient Methods for Accurate Sparse Trajectory Recovery and Map Matching](#). **ICDE 2025**.
- [2] **Wei Tian**, Jieming Shi, Siqiang Luo, Hui Li, Xike Xie, Yuanhang Zou. [Effective and Efficient Route Planning Using Historical Trajectories on Road Networks](#). **PVLDB 2023**, in the Proceedings of the VLDB Endowment, 2023.
- [3] Hai Lan, Jiong Xie, Zhifeng Bao, Feifei Li, **Wei Tian**, Fang Wang, Sheng Wang, Ailin Zhang. [VRE: A Versatile, Robust, and Economical Trajectory Data System](#). **PVLDB 2022**, 15(12): 3398-3410 (2022).
- [4] Di Peng, **Wei Tian**, Min Zhu, Yukun Ren, Xiaojian Lin, Mingzhao Li. [TargetingVis: Visual Exploration and Analysis of Targeted Advertising Data](#). *Journal of Visualization*, 2020, 23: 1113-1127.
- [5] Di Peng, **Wei Tian**, Binbin Lu, Min Zhu. [DMNEVis: A Novel Visual Approach to Explore Evolution of Dynamic Multivariate Network](#). *IEEE International Conference on Systems, Man, and Cybernetics (SMC)*, 2018, pp. 4304-4311.
- [6] Min Zhu, **Wei Tian**, Di Peng, Yabo Su, Hao Niu. [Method of Predicting User Professionalism Based on Question Granularity in Community Question Answering](#). *Advanced Engineering Sciences*, 2019, 51(1): 173-180.
- [7] Min Zhu, Jianhua Wang, Xiaowei Li, **Wei Tian**. [Fast Percentage Closer Soft Shadows](#). *Advanced Engineering Sciences*, 2019, 51(4): 140-146.
- [8] Min Zhu, Li Chen, Qin Li, **Wei Tian**, et.al. [Book: Handbook of Computer Network Course Projects \(in Chinese: 计算机网络课程设计\)](#). *China Machine Press (机械工业出版社)*, ISBN: 978-7-111-61139-4, 2018.
- [9] 朱敏, 彭第, 韦东鑫, **田伟**, 李亚男, 任宇堃, 林晓健. 层次数据的关联关系可视化处理方法、服务器及存储介质: 中国, 201911287989.8[P].
- [10] 朱敏, 李亚男, 彭第, 游思兰, 韦东鑫, **田伟**. 一种气泡层次可视化中气泡分离方法: 中国, 201811300412.1[P].

RESEARCH EXPERIENCE

- Aug. 2020 – Jun. 2021 *NoSQL Storage Engine for Big Trajectory Data Management and Analysis*
Research Intern in Database and Storage Laboratory, **Alibaba DAMO Academy**.
- ❖ Propose a Segment-based model to storing massive trajectory data, which saves about 3x – 4x storage cost.
 - ❖ Based on the Segment-based model, make the query performance of five basic categories 2x faster than existing methods.
 - ❖ Based on HBase, develop a storage engine for trajectory data.
- Aug. 2019 – Dec. 2019 *Research on visualization recommendation for relational data based on visual analytical task*
ISAR (Information Storage, Analysis and Retrieval) Research Group, **RMIT University**.
Visiting Student **RMIT Supervisor:** Associate Professor Zhifeng Bao
- ❖ Propose an end-to-end joint embedding learning framework, which aims to learn the association between Data-Task and visualization Views.
 - ❖ Propose a recommendation algorithm for visualization by fusing dataset features, analytical task features and visualization features with attention mechanism.

<i>Jan. 2019 – Apr. 2019</i>	<i>A Carpooling Algorithm Based on Region Division Using Trajectory Data</i> ❖ Propose a dividing urban area algorithm based on trajectory data by considering traffic conditions and semantic information in sub-regions. ❖ Propose a new carpooling model based on sub-regions by pruning the search space of matching riders and vehicles in region-level.
<i>May 2018 – May 2019</i>	<i>Research on selection pattern of advertising targets based on visual analysis</i> Project supported by Tencent Rhino Bird Program (Grant No. 18H0487) (¥200,000) Research Assistant ❖ Model the advertising behaviors of advertisers with multiple dimensions, such as industry type and product type. ❖ Analyze effect and relationship of targets, discovering classic cases and patterns in targets system.
<i>Jan. 2018 – Dec. 2019</i>	<i>Visual recommendation system based on user-generated content of tourism</i> Project supported by Sichuan Science and Technology Program of China (Grant No. 2018GZ0171) (¥200,000) Research Assistant ❖ Mine the frequent patterns of travel sequences from the massive travel blog contents by using NLP methods. ❖ Detect tourism area and popular local trip route.
<i>Feb. 2018 – May 2018</i>	<i>Visual Approach to Explore Evolution of Dynamic Multivariate Network</i> ❖ Propose a network visualization method based on community partitioning and hierarchical progress (group network exploration). ❖ Design a vertical stacking and adjustable network visualization method (ego-centric network exploration)
<i>Oct. 2017 – Jan. 2018</i>	<i>Expert Recommendation in Community Question Answering</i> ❖ Classify expertise metrics into community-level, topic-level and question-level, then propose a question-level expertise modeling method. ❖ Propose an expert recommendation algorithm by incorporating authority, interest and expertise.

PROFESSIONAL ACTIVITIES

<i>Teaching Assistant</i>	COMP4434, Big Data Analytics, Spring 2023, Hong Kong Polytechnic University
<i>Teaching Assistant</i>	COMP5434, Big Data Computing, Fall 2022, Hong Kong Polytechnic University
<i>Teaching Assistant</i>	COMP5434, Big Data Computing, Spring 2022, Hong Kong Polytechnic University
<i>Teaching Assistant</i>	ENG2002, Computer Programming, Fall 2021, Hong Kong Polytechnic University
<i>Invited Talks</i>	Jun. 2021: Vision Computing Lab, Sichuan University, China
<i>Teaching Assistant</i>	Urban Computing, Summer Term 2019, Sichuan University, China
<i>Teaching Assistant</i>	Big Data Management, Summer Term 2018, Sichuan University, China
<i>Teaching Assistant</i>	Course of Computer Network, Fall 2017, Sichuan University, China
<i>External Reviewers</i>	IEEE International Conference on Systems, Man, and Cybernetics (IEEE SMC'2018)
<i>Membership</i>	China Computer Federation Student Member

REPRESENTATIVE AWARDS

- The Hong Kong Polytechnic University Research Postgraduate Scholarship (2021)
- Outstanding Master Graduates of Sichuan University (2020)
- First-class of Excellent Postgraduate Scholarship by Sichuan University (2019)
- Outstanding Postgraduate of Sichuan University (2019)
- HUAWEI Scholarship of Sichuan University (2018)
- Outstanding student member of Sichuan University Branch of CCF (2018)
- Honorable mention of ChinaVis Data Challenge (Qingdao'2017)
- Outstanding Student of Sichuan University (2016)
- 2×National Encouragement Scholarship (2014, 2016)
- 2×Third-class of Excellent Undergraduate Scholarship by Sichuan University (9%) (2014, 2016)
- Second Prize of Sichuan Regional Contest of 7th Lan Qiao Cup (2016)